MCV Milling Vision
GUI User Manual

KUKA KR120 R2100 Fiber Milling Calibration System

Automated Vision-Based Robot Calibration

Version 1.0
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Django Web Application with Bootstrap 5 Dark Theme
ArUco Tag Detection | Hand-Eye Calibration | KRL Program Management

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1. Introduction

1.1 System Overview

MCV Milling Vision is a web-based application for automated vision-based calibration of the KUKA KR120 R2100 industrial robot used in fiber milling operations. The system uses ArUco marker detection to compute real-time corrections to the robot's base frame, ensuring precise part positioning throughout production cycles.

The application provides:

- Real-time robot position monitoring and jog control
- ArUco tag-based part and table detection
- Automated calibration cycle with correction computation
- Hand-eye calibration for camera-to-robot transformation
- Live camera feed via WebSocket streaming
- KRL program upload, management, and execution
- Intrinsic camera calibration via chessboard pattern
- 2D/3D workspace visualization
- Event logging and calibration history export (CSV/Excel)
- Full settings backup and restore

1.2 System Requirements

Component	Requirement
Robot	KUKA KR120 R2100 with EKI interface
Camera	USB camera (recommended: industrial grade)
Network	TCP/IP connection to robot controller
Browser	Chrome, Firefox, or Edge (modern version)
Server	Python 3.10+, Django 5.x
Ports	8000 (HTTP), 8001 (WebSocket/Daphne)

1.3 Starting the Application

To start the MCV Vision system, two servers must be launched:

Step 1: Open a terminal and navigate to the project directory.

```
cd mcv_milling
```

Step 2: Start the Django development server (HTTP on port 8000):

```
python manage.py runserver 0.0.0.0:8000
```

Step 3: In a second terminal, start Daphne for WebSocket support (port 8001):

```
daphne -b 0.0.0.0 -p 8001 mcv_milling.asgi:application
```

Step 4: Open your browser and navigate to:

```
http://localhost:8000
```

Server	Port	Purpose
Django (runserver)	8000	HTTP pages, REST API, static files
Daphne (ASGI)	8001	WebSocket for live camera streaming

Note: The Daphne server is only required for the live camera feed. All other features work with just the Django server.

2. Navigation & Layout

The application uses a fixed sidebar layout with a top bar. All pages share the same navigation structure for consistent user experience.

2.1 Sidebar Navigation

The left sidebar is always visible and organizes navigation into four logical sections:

Section	Page	Description
OVERVIEW	Dashboard	System status, camera feed, event log
OPERATIONS	Calibration	Setup, control, run calibration cycles
OPERATIONS	Jobs	Upload and execute KRL programs
VISION	Camera Cal	Intrinsic camera calibration
SYSTEM	Settings	All system configuration parameters

The sidebar footer displays the real-time robot connection status with a colored indicator (green = connected, red = disconnected). The active page is highlighted in the sidebar.

2.2 Top Bar

The top bar spans across the main content area and displays:

- Page title (left side)
- Theme toggle button (sun/moon icon)
- Robot name badge (shows configured robot name or '--')
- Connection status badge ('Connected' in green or 'Disconnected' in red)
- User info and logout button (when authenticated)

2.3 Theme Toggle

Click the sun/moon icon in the top bar to switch between dark and light themes. Your preference is saved automatically and persists across browser sessions.

2.4 Toast Notifications

The system displays brief notification messages (toasts) in the bottom-right corner of the screen whenever an action is performed. Toasts are color-coded by severity:

- **Info:** Blue - Information messages
- **Success:** Green - Successful operations
- **Warning:** Yellow - Warnings requiring attention
- **Error:** Red - Error messages (displayed for 6 seconds)

3. Getting Started Wizard

The Getting Started wizard is displayed at the top of the Dashboard page. It provides a visual, step-by-step guide that shows the current system readiness and walks you through the initial setup process. Each step shows its current status and links to the relevant page.

The wizard displays a horizontal progress bar that fills as steps are completed. Steps are shown with colored icons: green (completed), orange (active/current), or gray (locked/pending).

3.1 Step 1: Configure Settings

Ensure the robot IP address and port are configured in the Settings page. The wizard checks that both `robot_ip` and `robot_port` fields are set. Click this step to navigate to Settings.

- Status shows 'Configured' (green) when both fields have values
- Status shows 'Not set' (red) when either field is empty

3.2 Step 2: Connect Robot

Establish a TCP connection to the KUKA robot controller via the EKI interface. Click this step to focus the robot IP input field on the Dashboard.

- Status shows 'Connected' (green) when the robot is online
- Status shows 'Not connected' (red) when disconnected

3.3 Step 3: Open Camera

The camera must be opened and streaming. Click this step to navigate to the Calibration page where you can start the camera feed.

- Status shows 'Opened' (green) when the camera is active
- Status shows 'Not open' (red) when the camera is closed

3.4 Step 4: Teach Nominal Position

The nominal (reference) part position must be taught before calibration can run. Click this step to navigate to the Calibration page's Calibrate tab.

- Status shows 'Taught' (green) when the nominal position is saved
- Status shows 'Not taught' (red) when no nominal exists

3.5 Step 5: Calibrate!

The final step is unlocked only when all four prerequisite steps are complete. It shows how many steps remain, or 'Ready!' when all prerequisites are met.

> **TIP:** The wizard automatically polls the system status every time you open the Dashboard. Complete steps in any order - the wizard updates in real-time.

4. Dashboard

The Dashboard is the main overview page providing a comprehensive view of system status, live camera feed, statistics, calibration history, and event logging.

4.1 Robot Connection

The Robot Connection card allows you to connect or disconnect from the KUKA robot controller.

- **IP & Port:** Enter the robot's IP address and EKI port number
- **Connect:** Click 'Connect' (green button) to establish the TCP connection
- **Disconnect:** Click 'Disconnect' (red button) to close the connection

Note: The connection status is polled every 5 seconds and reflected in the sidebar, top bar, and all relevant pages across the application.

4.2 Live Camera Feed

The camera feed card displays a real-time video stream from the connected camera via WebSocket.

- **Play:** Click the play button (green) to start the video stream
- **Stop:** Click the stop button (red) to stop streaming

WARNING: The live camera feed requires the Daphne ASGI server running on port 8001. Without it, the feed will not work.

4.3 Statistics

Displays quick statistics for the current session:

- Total Cycles - number of calibration cycles run
- Successful - number of successful calibrations
- Active Program - currently loaded KRL program name

4.4 Calibration History Chart

A line chart (Chart.js) showing the history of calibration corrections over time. Three lines are plotted: dX (red), dY (blue), and dZ (teal), measured in millimeters. Click the refresh button to reload the chart data.

4.5 Event Log

A scrollable table showing the most recent system events, auto-refreshing every 10 seconds.

- **Filter:** Filter events by category using the dropdown: Robot, Calibration, Camera, Job, Settings, System
- **Levels:** Events are color-coded: info (blue), success (green), warning (yellow), error (red)
- **Refresh:** Click refresh to manually reload the event list

4.6 Recent Calibrations Table

A detailed table of recent calibration results showing per-axis corrections (dX, dY, dZ, dA, dB, dC), translation and rotation magnitudes, and status (OK, FAIL, REJ).

- **CSV Export:** Click 'CSV' to export calibration data as a CSV file
- **Excel Export:** Click 'Excel' to export as an XLSX spreadsheet

5. Calibration Page

The Calibration page is the core operational hub of the system. It is organized into five tabs to group related functions and eliminate scrolling. Each tab focuses on a specific phase of the calibration workflow.

Tab	Icon	Purpose
Setup	Check-square	Prerequisites check, tag scanning, position readout
Control	Joystick	Robot jog panel, capture position
Calibrate	Crosshair	Teach nominal, run calibration, auto-cal
Vision	Camera	Camera feed, 2D workspace, segmentation
Advanced	Gear	3D visualization, hand-eye cal, HSV tuner

5.1 Setup Tab

5.1.1 Prerequisites Checklist

Displays a real-time readiness checklist showing whether each system component is ready:

- Hand-Eye Calibration - whether a valid hand-eye matrix is loaded
- Nominal Position Taught - whether the reference position has been saved
- Camera Connected - live status, polled every 5 seconds
- Robot Connected - live status, polled every 5 seconds

5.1.2 Tag Scanner

The Tag Scanner detects all visible ArUco markers in the camera frame and lets you assign them roles (Table tag or Part tag).

Step 1: Click 'Scan All Tags' to detect visible markers.

Step 2: Review detected tags showing ID and distance.

Step 3: Use the dropdown for each tag to assign: Table or Part.

Step 4: Click 'Save Assignment' when exactly 1 Table and 1 Part tag are selected.

Note: Current tag assignments are displayed above the scan results.

5.1.3 Robot Position

Shows the current robot TCP position in both Cartesian (X, Y, Z, A, B, C) and Joint (A1-A6) coordinates.

- **Live Mode:** Click 'Live' to enable continuous position updates at 2 Hz
- **Manual Refresh:** Click 'Refresh' for a one-time position update

5.2 Control Tab

5.2.1 Position Readout

A compact horizontal bar showing real-time X, Y, Z, A, B, C values, mirrored from the Setup tab's robot position card.

5.2.2 Robot Jog Panel

Provides manual control to move the robot in small increments. Essential for positioning the robot during setup and teaching operations.

- **Mode:** Toggle between Cartesian (X/Y/Z/A/B/C) and Joint (A1-A6) jog modes
- **Step Size:** Select step size: 0.1, 1, 10, or 50 mm (or degrees for rotation)
- **Speed:** Adjust speed override: 1-100% (default 10%)
- **Jog Buttons:** Press +/- buttons for each axis to jog the robot

CAUTION: Always start with small step sizes (0.1 or 1 mm) and low speed. Larger steps can cause rapid robot movement. Ensure the workspace is clear before jogging.

5.2.3 Capture Position

Saves the current robot joint position as the capture (measurement) position. This is the pose the robot moves to when performing a calibration cycle to take camera images.

- Jog the robot to the desired capture position with the camera viewing the workspace
- Click 'Teach from Current' and confirm the dialog to save

5.3 Calibrate Tab

5.3.1 Step 1: Teach Nominal Position

The nominal position is the reference location of the part relative to the table tag. Place the part at its ideal position with the ArUco tag mounted, then click 'Teach Nominal'.

WARNING: Both the table tag and part tag must be visible to the camera at the capture position. If either tag is not detected, the teach operation will fail.

5.3.2 Step 2: Run Calibration Cycle

Click the large green 'CALIBRATE' button to run a single calibration cycle. The system will:

- Move the robot to the capture position
- Capture a camera image and detect ArUco tags
- Compute the displacement from the nominal position
- Calculate the corrected base frame ($BASE)
- Send the correction to the robot controller

The Calibration Result card appears showing per-axis corrections (dX-dC) and the corrected base frame values.

5.3.3 Auto-Calibration

Run multiple calibration cycles automatically:

- **Cycles:** Set the number of cycles (1-20, default 3)
- **Auto-Calibrate:** Click 'Auto-Calibrate' to run N consecutive cycles
- **Repeatability:** Click 'Repeatability' to run a repeatability test that reports statistics (mean, std, range)
- **Dry Run:** Enable the 'Dry Run' switch to simulate without sending corrections to the robot

Results show average corrections per axis, color-coded: green (below 1mm), yellow (1-5mm), red (above 5mm).

5.4 Vision Tab

5.4.1 Camera Feed

Live camera stream displayed via WebSocket, identical to the dashboard camera feed. Use play/stop buttons to control streaming.

5.4.2 XY Plane Workspace Visualization

A 2D canvas rendering showing a bird's-eye view of the workspace in the table tag's coordinate frame.

- **Table Tag:** Table tag shown at the origin as a green square
- **Nominal:** Nominal part position shown as a cyan wireframe
- **Current:** Current part position shown as an orange circle
- **Displacement:** Red dashed arrow shows displacement from nominal to current

- Click 'Capture' for a one-time workspace snapshot
- Click 'Auto' to continuously update every second

5.4.3 Workspace Coordinates

Numerical coordinate readout showing table tag origin, current part position, nominal position, and computed displacement. Displacement values are color-coded by magnitude.

5.4.4 Segmentation Diagnostics

Shows a 4-panel diagnostic view of the tag detection pipeline: Raw image, CLAHE enhanced, adaptive threshold, and detection overlay. Useful for debugging detection issues.

- Click 'Capture' for a one-time diagnostic frame
- Click 'Auto' for continuous diagnostics at 1.5-second intervals
- Shows number of tags found, rejected candidates, and per-tag details

5.5 Advanced Tab

5.5.1 3D Workspace View

An interactive Three.js 3D scene showing the spatial relationship between the table tag, nominal position, and current part position. Use the mouse to orbit, zoom, and pan.

- Green cube at origin = table tag
- Blue wireframe = nominal position
- Orange sphere = current part position
- Click 'Update' to refresh positions from the camera

5.5.2 Hand-Eye Calibration

Computes the transformation matrix between the camera and the robot's end-effector (hand-eye calibration). This is required for accurate coordinate transformation.

Step 1: Move the robot so the table tag is visible from different angles.

Step 2: Click 'Capture Sample' at each pose (minimum 5 samples required).

Step 3: Click 'Compute' when you have enough samples.

Step 4: Click 'Apply to Settings' to save the computed matrix.

TIP: Capture samples from widely varying angles and distances. More diverse poses yield a more accurate hand-eye calibration. 10-15 samples from different orientations is recommended.

- **Reset:** Click 'Reset' (red) to clear all samples and start over

5.5.3 HSV Color Tuner

Adjust HSV (Hue, Saturation, Value) color thresholds for image preprocessing. Six sliders control the lower and upper bounds for H, S, and V channels.

- Adjust sliders to define the color range of interest
- Click 'Preview' to see the filtered camera image
- The mask percentage shows how much of the image falls within the HSV range
- Click 'Save' to store the HSV bounds to settings

6. KRL Programs (Jobs)

The Jobs page manages KUKA Robot Language (KRL) programs. You can upload .src/.dat files, view source code with syntax highlighting, activate programs, and execute calibrated motion cycles.

6.1 Uploading Programs

Step 1: Select a **.src file** (required) - the main KRL source.

Step 2: Optionally select a **.dat file** - the data file with point declarations.

Step 3: Enter a program name (or leave blank to auto-detect from the DEF statement).

Step 4: Optionally add a description.

Step 5: Click 'Upload' to submit.

Note: The system automatically parses the .src file to count motion points.

6.2 Managing Programs

The Programs table lists all uploaded KRL programs with the following actions:

Action	Icon	Description
View Source	Code	Opens a modal with syntax-highlighted KRL code
Activate	Check	Sets this program as the active program for execution
Statistics	Chart	Shows execution statistics (cycles, errors, durations)
Delete	Trash	Permanently removes the program (with confirmation)

The source viewer provides syntax highlighting for KRL keywords (DEF, END), motion commands (PTP, LIN, CIRC), control flow (IF, FOR, WHILE), and comments.

6.3 Executing Cycles

The Execute Cycle card runs the active KRL program on the robot:

- Enable 'Auto-calibrate before execution' (checked by default) to run a calibration cycle before program execution
- Click 'Execute Cycle' to start - the robot will run the loaded program
- Results show cycle number, duration, and whether calibration was applied

CAUTION: Ensure the workspace is clear and safety systems are active before executing robot programs.

7. Camera Calibration

The Camera Calibration page performs intrinsic camera calibration using a chessboard pattern. This calibration determines the camera's focal length, principal point, and distortion coefficients, which are essential for accurate ArUco tag detection.

7.1 Chessboard Settings

Configure the chessboard pattern parameters:

Parameter	Default	Description
Rows	9	Number of inner corners (rows)
Cols	6	Number of inner corners (columns)
Square (mm)	25	Physical size of each square in millimeters

Note: The rows and columns refer to inner corners, not squares. A standard 10x7 chessboard has 9x6 inner corners.

7.2 Capturing Frames

Step 1: Hold the chessboard pattern in front of the camera.

Step 2: Click 'Detect Board' to preview corner detection (without saving).

Step 3: When corners are detected correctly, click 'Capture Frame' to save.

Step 4: Move the board to a different angle/distance and repeat.

Step 5: Capture at least 5 frames (15-25 recommended for best results).

A progress bar shows how many frames have been captured out of the recommended 15.

TIP: Vary the board position, angle, and distance between captures. Include tilted views and positions near the edges of the frame for better calibration.

7.3 Computing Calibration

Once 5 or more frames are captured, the 'Compute Calibration' button becomes active. Click it to compute the camera intrinsic parameters. The results show:

- RMS Error - reprojection error (green: below 1, yellow: 1-2, red: above 2)
- fx, fy - focal lengths in pixels
- cx, cy - principal point coordinates
- Frames - number of frames used in computation

Click 'Apply to Settings' to save the computed parameters as the active camera calibration. The page will reload to reflect the new values.

7.4 Calibration History

A table of all previous camera calibrations showing date, RMS error, pose count, intrinsic parameters, board dimensions, and whether the calibration is currently applied. You can apply any historical calibration by clicking the 'Apply' button in its row.

8. Settings

The Settings page centralizes all system configuration. Changes are saved by clicking the green 'Save Settings' button at the bottom of the form.

8.1 Robot Configuration

Field	Description
IP Address	KUKA robot controller IP (e.g., 192.168.1.1)
Port	EKI communication port (e.g., 54610)
Override %	Robot speed override percentage (1-100%)

8.2 Camera Configuration

Field	Description
Index	Camera device index (usually 0)
FPS	Desired frames per second
Width / Height	Camera resolution in pixels
fx, fy	Focal lengths (from camera calibration)
cx, cy	Principal point (from camera calibration)

8.3 ArUco Tag Configuration

Field	Description
Dictionary	ArUco dictionary type (auto-detected if wrong)
Table Tag ID	ArUco ID of the table reference tag
Part Tag ID	ArUco ID of the part tag
Tag Size (mm)	Physical size of the ArUco marker
Max Correction (mm)	Maximum allowed translation correction
Max Correction (deg)	Maximum allowed rotation correction

Note: Corrections exceeding the maximum thresholds will be rejected (status: REJ) to prevent unsafe robot movements.

8.4 Nominal Base Frame

The nominal $BASE frame defines the robot's reference coordinate system. Enter the six components: X, Y, Z (mm) and A, B, C (degrees). This frame is the starting point for calibration corrections.

8.5 Hand-Eye Matrix

The 4x4 transformation matrix from hand-eye calibration. This can be pasted as a JSON array or computed and applied from the Calibration > Advanced > Hand-Eye Calibration card.

8.6 Base Frame Management

Manage multiple robot base frames (up to 32, matching KUKA $BASE[1] through $BASE[32]).

- **Add:** Click 'Add Frame' to create a new base frame with name, BASE number, and X/Y/Z/A/B/C values
- **Activate:** Click 'Activate' to set a frame as the active calibration reference
- **Edit:** Click 'Edit' to modify an existing frame's values
- **Delete:** Click 'Delete' to remove a frame (with confirmation)

Each frame shows badges: 'HE' (blue) if hand-eye data is attached, 'NOM' (yellow) if nominal data is saved for that frame.

8.7 Backup & Restore

Create a complete backup of all settings, base frames, and capture positions as a JSON file.

- **Backup:** Click 'Download Backup' to save a JSON backup file
- **Restore:** Click 'Restore from File' to upload and apply a previous backup

WARNING: Restoring from a backup will overwrite ALL current settings. A confirmation dialog will appear before the restore is applied.

9. Troubleshooting

Robot won't connect

Verify the IP address and port match the KUKA controller's EKI configuration. Ensure the robot is powered on and the network cable is connected. Check that no firewall is blocking the port.

Camera feed is black / won't start

Ensure the Daphne ASGI server is running on port 8001. Check that the camera index in Settings matches your camera device. Try a different camera index (0, 1, 2).

Tags not detected

Ensure adequate lighting. Check that the correct ArUco dictionary is selected in Settings. Use the Segmentation Diagnostics in the Vision tab to debug detection. Ensure tags are not too far from the camera or at extreme angles.

Calibration returns REJ status

The computed correction exceeds the maximum thresholds set in Settings. Either increase the Max Correction limits or investigate why the part displacement is so large.

Teach Nominal fails

Both the table tag and part tag must be visible to the camera at the capture position. Check tag assignments in the Setup tab's Tag Scanner.

Hand-Eye calibration is inaccurate

Capture more samples (15+) from diverse robot poses. Ensure the table tag is fully visible in all captured frames. Avoid capturing samples where the tag is at the edge of the camera frame.

Camera calibration RMS is high (> 2.0)

Re-capture calibration frames. Ensure the chessboard is flat and well-lit. Include frames from various angles and distances. Avoid motion blur.

WebSocket connection fails

Ensure the Daphne server is running: daphne -b 0.0.0.0 -p 8001 mcv_milling.asgi:application. Check browser console for WebSocket errors.

10. Quick Reference

10.1 Typical Calibration Workflow

Step	Action	Page / Tab
1	Configure robot IP, port, and camera settings	Settings
2	Connect to the robot	Dashboard
3	Start the camera feed and verify image quality	Dashboard or Calibration > Vision
4	Run Camera Calibration (if not done)	Camera Cal
5	Perform Hand-Eye Calibration (if not done)	Calibration > Advanced
6	Scan and assign Table/Part tags	Calibration > Setup
7	Jog robot to capture position and save it	Calibration > Control
8	Place part at reference position, Teach Nominal	Calibration > Calibrate
9	Run calibration cycle(s)	Calibration > Calibrate
10	Verify results in Vision tab and History chart	Calibration > Vision / Dashboard

10.2 Color Code Reference

Color	Meaning
Green	Connected / OK / Completed / Below threshold
Orange	Active step / Current / Accent highlights
Yellow	Warning / Part tag / 1-5mm correction
Red	Error / Disconnected / Above 5mm correction
Cyan	Nominal values / Joint positions / Info
Blue	Information / Hand-eye data / Interactive elements

10.3 Calibration Status Codes

Code	Badge	Meaning
OK	Green	Calibration successful, correction applied to robot
FAIL	Red	Calibration failed (tag detection error, computation error)
REJ	Yellow	Correction rejected (exceeded max threshold limits)